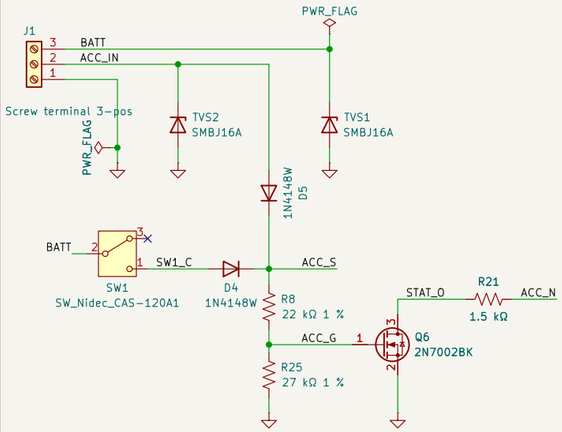
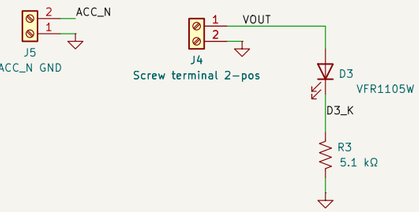
INPUT / ACC SENSE

Harness fuse; SW1 closed = key on



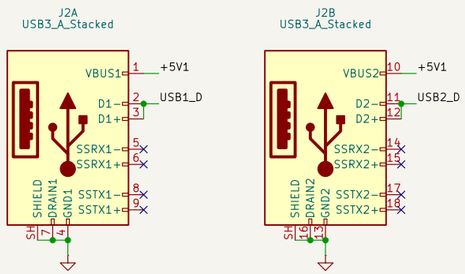
STATUS / LED / PI 12 V

J5 low = ACC on (Pi pull-up)



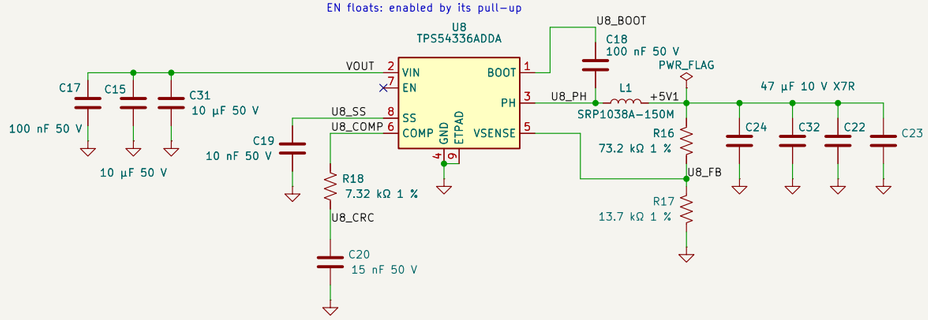
USB

D+ shorted to D− per port (BC1.2)



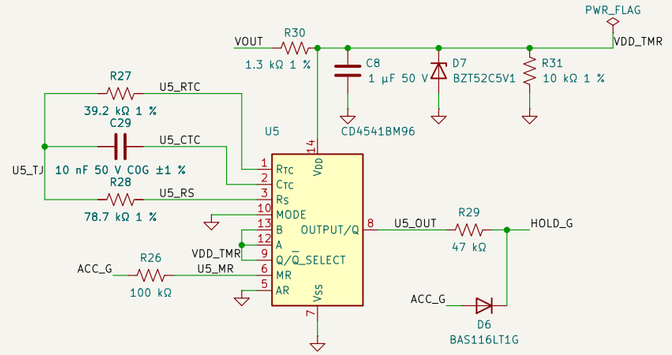
BUCK 5.07 V

EN floats: enabled by its pull-up



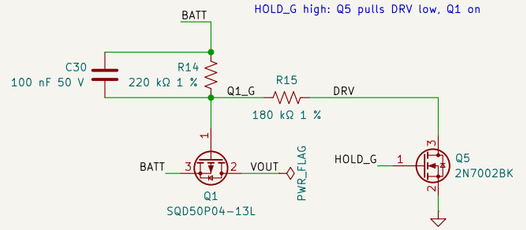
TIMER U5

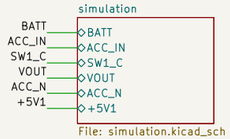
ACC holds reset; key off: OUT low after the count; U5 runs from VOUT



P-FET SWITCH

HOLD_G high: Q5 pulls DRV low, Q1 on



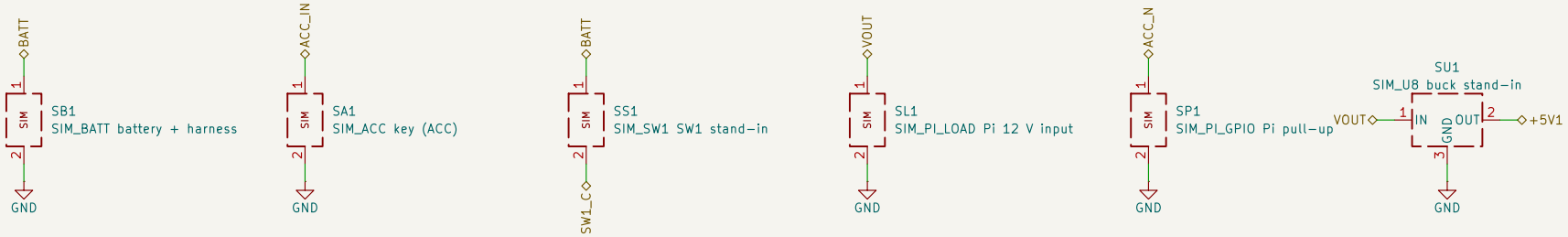


Sheet: /
File: pi_power_board.kicad_sch

Title: Pi 5 car power HAT

Size: B	Date:	Rev:
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SIM-ONLY SHEET: not on the board or in the BOM. Inspect > Simulator: FAST default (U5 count / U5_TAPDIV). pdm run sim; --full: real time (tools/run_sim.py, reports/sim_kicad.md)




```
* SIM-ONLY (tools/post_sch/12, 13): defaults = the FAST nominal key cycle at 12 V, one key cycle
* FAST: U5 counts 2^15 / U5_TAPDIV clocks on the real R27 / C29 / R28; hold_est adds the rest at the measured period.
* Real time: U5_TAPDIV=1, T_PIOFF / T_CW1 unscaled, T_STOP past the hold, .options interp (pdm run sim --full)
* corners (tools/run_sim.py): R27 / C29 factors, U5 oscillator, diodes, FET V_GS(th) and C_gd, zener, U5 current
.param RFAC=1 CFAC=1 C8_RET=0.7
.param U5_HYS=0.054 U5_RDRV=129 U5_IDD=0.3m U5_TAPDIV=256 N_CNT=32768
.param IS_D=1.5e-10 BVZ=5.1 VTO_N=1.6 IDSS_N=0 VTO_P=-2.0 IDSS_P=0 CGD_FAC=1
* battery, key, SW1
.param VB=12 RHARN=50m LHARN=1u T_PLUG=10m TEDGE=1m
.param ACC_V=12 T_ON1=1 T_OFF1=5 T_ON2=1e3 T_OFF2=1.001e3
.param SW_EN=0 T_SW_ON=1 T_SW_OFF=5
* Pi load (constant power) and GPIO pull-up
.param CVOUT=674u VUV=4.5 P_RUN=27 P_AFTER=2 T_PIOFF=5.0117 T_PISD=3 VPI=3.3 RPI=50k
* U8 stand-in (CIN_U8: C15 + C17 effective), USB load, overload
.param V5SET=5.07 UV_ON=4.5 UV_OFF=4.32 T_SS=3.5m C5V1=64u CIN_U8=10u RFB=86.9k I_USB=3 ETA=0.91 ILIM=4.7
.param OVL=0 R_OVL=0.5 T_OVL=13 T_HWAIT=1.255m T_HOFF=40.16m
* windows after the last ACC / SW1 falling edge T_LASTOFF: counting T_CW0 to T_CW1, VOUT on from T_VON0;
* U5 period between the CTC rising edges N_P1 and N_P2; logic threshold VTH_MEAS
.param T_LASTOFF=5 T_CW0=20m T_CW1=78.1m T_VON0=1.5 N_P1=16 N_P2=80 VTH_MEAS=2.5
* output grid T_STEP (with interp); T_END = last point; keep T_STEP / T_STOP / T_MAX equal to the .tran numbers
.param T_STOP=6 T_STEP=1m T_MAX=1m T_ENDWIN=0.3
.param T_END={T_STOP-2*T_STEP}
```

```
.options method=gear
.save V(/BATT) V(/ACC_IN) V(/ACC_G) V(/HOLD_G) V(/VOUT) V(/VDD_TMR) V(/U5_OUT) V(/ACC_N) V(/+5V1) V(/U5_CTC)
.tran 1m 6 0 1m uic
* hold: last ACC / SW1 falling edge to VOUT falling through VB / 2
.meas TRAN vout_fall FIND time WHEN V(/VOUT)={VB/2} TD={T_LASTOFF} FALL=1
.meas TRAN hold PARAM='vout_fall - T_LASTOFF'
.meas TRAN vout_fall_last FIND time WHEN V(/VOUT)={VB/2} FALL=LAST
.meas TRAN vout_rise_last FIND time WHEN V(/VOUT)={VB/2} RISE=LAST
* U5: end of the count, oscillator period on CTC (raw points: no interp), the hold with all N_CNT counts
.meas TRAN u5_fall FIND time WHEN V(/U5_OUT)={VTH_MEAS} TD={T_LASTOFF} FALL=1
.meas TRAN osc_t1 FIND time WHEN V(/U5_CTC)={VTH_MEAS} TD={T_LASTOFF} RISE={N_P1}
.meas TRAN osc_t2 FIND time WHEN V(/U5_CTC)={VTH_MEAS} TD={T_LASTOFF} RISE={N_P2}
.meas TRAN t_osc PARAM='(osc_t2 - osc_t1)/(N_P2 - N_P1)'
.meas TRAN hold_est PARAM='hold + (N_CNT - N_CNT/U5_TAPDIV)*t_osc'
* on / counting: VOUT never dips, VDD_TMR range, HOLD_G, J5
.meas TRAN vout_min_on MIN V(/VOUT) FROM={T_VON0} TO={T_LASTOFF+T_CW1}
.meas TRAN vtmr_cnt_min MIN V(/VDD_TMR) FROM={T_LASTOFF+T_CW0} TO={T_LASTOFF+T_CW1}
.meas TRAN vtmr_cnt_max MAX V(/VDD_TMR) FROM={T_LASTOFF+T_CW0} TO={T_LASTOFF+T_CW1}
.meas TRAN holdg_cnt_min MIN V(/HOLD_G) FROM={T_LASTOFF+T_CW0} TO={T_LASTOFF+T_CW1}
.meas TRAN j5_on_max MAX V(/ACC_N) FROM={T_ON1+0.5} TO={T_OFF1}
.meas TRAN j5_off_min MIN V(/ACC_N) FROM={T_LASTOFF+T_CW0} TO={T_LASTOFF+T_CW1}
* end: VOUT / VDD_TMR / HOLD_G off in the last T_ENDWIN, no re-latch
.meas TRAN vout_end_max MAX V(/VOUT) FROM={T_STOP-T_ENDWIN} TO={T_END}
.meas TRAN vtmr_end_max MAX V(/VDD_TMR) FROM={T_STOP-T_ENDWIN} TO={T_END}
.meas TRAN holdg_end_max MAX V(/HOLD_G) FROM={T_STOP-T_ENDWIN} TO={T_END}
* battery hot-plug, key off: VOUT, VDD_TMR, HOLD_G before the first key-on
.meas TRAN hp_vout_max MAX V(/VOUT) FROM=0 TO={min(min(T_ON1, T_SW_ON)-TEDGE, T_END)}
.meas TRAN hp_vtmr_max MAX V(/VDD_TMR) FROM=0 TO={min(min(T_ON1, T_SW_ON)-TEDGE, T_END)}
.meas TRAN hp_holdg_max MAX V(/HOLD_G) FROM=0 TO={min(min(T_ON1, T_SW_ON)-TEDGE, T_END)}
```

Sheet: /simulation/
File: simulation.kicad_sch

Title: Pi 5 car power HAT: simulation (sim-only)

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